
When Do Transformer Components Compose?

Validating a Log-Pool Decomposition Criterion

Anonymous¹

Abstract

A recent log-pool framework introduces a compositional utility gap as a criterion for decomposing a probability distribution into interacting contributions. We ask whether this gap, applied to architectural decompositions of trained transformers, has empirical content independent of any sub-agent interpretation. We measure the gap at three resolutions: atomic per-component gaps, random partitions, and partitions found by a genetic algorithm. A partition is feasible when the smallest group gap is non-negative. Held-out recomputation is stable; cross-corpus transfer depends on resolution and architecture. Across Llama 3.1 8B, Pythia 1.4B, and Gemma 2 2B, random-partition feasibility differs sharply under three nulls (uniform, size-matched, residual-matched): Llama is permissive, Pythia rare, Gemma zero in every cell; the genetic algorithm finds a feasible partition in all but one trained cell. Under per-tensor- σ -matched parameter-scrambled controls, both random and search feasibility collapse in every cell. Under these stated conventions, the gap is a substantive measurement tool for learned transformer structure, while leaving open whether the measured property reflects subagent structure, distributional coherence, or broader decomposition coherence.

1. Validation Problem

The log-pool framework of Lee et al. (2025) defines a compositional utility gap

$$\Delta_S = \mathbb{E}_x \left[\sum_y (P(y|x) - P_S(y|x)) \log P_S(y|x) \right],$$

where P is the full-model next-token distribution and P_S is the distribution induced by subset S of the model’s archi-

tectural components, as a criterion for when S is compositionally informative. The original formalism is motivated by a latent-agentic picture, but the present paper asks a prior empirical question: when this criterion is applied to architectural decompositions of trained transformers, does it measure anything non-trivial?

This question matters because a criterion can be formally well-defined while empirically uninformative. If log-pool feasibility held equally on train and validation only by overfitting, failed to distinguish architecture and measurement settings, or survived per-tensor- σ -matched scrambling of learned weights, then it would be a weak measurement tool for mechanistic interpretation. Conversely, if the gap generalizes, discriminates across architecture and measurement settings, and separates trained models from parameter-scrambled controls, then the metric has empirical content independently of whether the stronger subagent interpretation is correct.

We therefore validate the gap along three axes:

1. *generalization*: train, validation, and cross-corpus behavior;
2. *model/protocol variation*: Llama 3.1 8B (RMSNorm, no soft-cap), Pythia 1.4B (LayerNorm), and Gemma 2 2B (RMSNorm with post-attention normalization and final logit soft-cap; we use pre-cap logits);
3. *trained-structure sensitivity*: trained models versus parameter-scrambled controls.

We probe each axis across three measurement resolutions: atomic per-component gaps, random-partition feasibility, and optimized-partition feasibility (cross-corpus transfer is reported only at the optimized level). The three architectures span the normalization and capping conventions that affect exact logit additivity in our protocols; the three resolutions span the aggregation levels available from a top- K component tensor. The contribution is a measurement-validation result: the compositional utility gap discriminates trained-transformer structure across generalization, model/protocol variation, and scrambled-control tests, without committing to a metaphysical theory of neural parts.

¹Anonymous Submission. Correspondence to: Anonymous <anon@anon>.

2. Setup

Architectural logit decomposition. For a transformer with N architecturally natural components, we decompose pre-softmax logits as

$$z(x) = \sum_{i=1}^N z_i(x),$$

where components are the embedding, each attention sublayer, and each MLP sublayer. Llama 3.1 8B has $N = 65$ components (Grattafiori et al., 2024); Pythia 1.4B has $N = 49$ (Biderman et al., 2023); Gemma 2 2B has $N = 53$ under the same layer-level convention (Gemma Team et al., 2024). Appendix A records the normalization and soft-cap conventions needed for exact additivity.

Log-pool feasibility. For a component set S , we define a group distribution $P_S = \text{softmax}(l_S)$ from a group logit l_S . For a partition $\Pi : \{1, \dots, N\} \rightarrow \{0, 1, \dots, r\}$, group 0 is residual and non-empty groups $j \in \{1, \dots, r\}$ are evaluated by their gaps Δ_j^Π . A partition is feasible when $\min_j \Delta_j^\Pi \geq 0$ over non-empty non-residual groups. Sums over y are approximated on top- $K = 10,000$ vocabulary tensors.

Measurement conventions. We use the gap functional under two stated logit-temperature conventions. Atomic singleton measurements use a down-scaled singleton diagnostic, $P_{\{i\}} = \text{softmax}(z_i/N)$, matching the atomic pipeline. Random-partition and GA partition measurements use the unnormalized product-of-experts group convention used in the partition experiments,

$$l_S = \sum_{i \in S} z_i, \quad P_S = \text{softmax}(l_S).$$

These partition results are not simplex-normalized log-pool reconstructions in the sense of Lee et al. (2025); they are product-of-experts group scores over additive architectural logits. Appendix B states the exact simplex-normalized reconstruction $\tilde{P}_S = \text{softmax}(z_S/\beta_S)$ separately. Atomic gap magnitudes and partition min-gaps are therefore not on the same temperature scale. Re-estimating partition results under simplex-normalized group temperatures is left for follow-up.

Measurement levels. We evaluate the gap at three levels.

- *Atomic:* the distribution of singleton gaps $\Delta_{\{i\}}$ across architectural components.
- *Random partitions:* Monte Carlo estimates of $\Pr(\text{feasible})$ under uniform-random, size-matched, and residual-matched partition nulls ($n = 1,000$ partitions per cell).

- *Optimized partitions:* whether a genetic algorithm can find feasible partitions, and the recomputed validation min-gap of the best found partition.

Parameter-scrambled control. The N1 control re-samples each weight tensor from $\mathcal{N}(0, \sigma_i^2)$ using the original tensor’s empirical standard deviation, sets norm gains to 1, and biases to 0. This preserves architecture, tensor scale, dtype conventions, and extraction code while destroying learned content. The control should be read as parameter-scrambled, not as a standard random initialization.

3. Three-Axis Validation

3.1. Axis 1: Generalization

The first validation axis asks whether the gap survives re-computation rather than only reflecting search-time artifacts. At the atomic level, singleton gap distributions are measured on held-out validation tensors; the per-cell distributions appear in Appendix G. At the random-partition level, all null-baseline rates in Table 4 are evaluated on validation tensors rather than training logs. At the optimized level, Table 2 reports cross-corpus transfer of trained GA partitions: each saved partition is re-evaluated on validation tensors from all source corpora. The pattern is resolution-dependent: $r=10$ transfer is broad (6 of 6 Llama transfers feasible; both Pythia code→wiki and wiki→code feasible), while $r=20$ transfer becomes selective (3 of 6 Llama transfers fail self-eval on the cross-corpus side; Pythia code→wiki fails). The full transfer table is in Appendix H.

3.2. Axis 2: Model/Protocol Variation

The second validation axis asks whether the gap distinguishes architecture and measurement settings. At level (b), Table 4 shows that random-partition feasibility is itself architecture-specific rather than a generic trained-transformer property: Llama at $r=10$ is permissive (50–93% rates under all three nulls), Pythia at $r=10$ is tight (≤ 0.15), and Gemma admits zero feasible random partitions across all six (domain, r) cells $\times$ three nulls $\times$ 1,000 samples. Llama at $r=20$ is intermediate and null-dependent. The size-matched random null is consistently more permissive than uniform-random, sometimes substantially (Llama code $r=10$: 0.722 $\rightarrow$ 0.932; Llama arxiv $r=20$: 0.257 $\rightarrow$ 0.562), indicating that partition geometry alone explains some of the feasibility before learned structure is invoked. At level (a), Table 3 shows per-component atomic gap magnitudes ordered *inversely* to the partition-level ranking: Gemma has the largest mean atomic gap ($\sim 1 \cdot 10^{-2}$), Pythia next ($\sim 4\text{--}6 \cdot 10^{-3}$), Llama smallest ($\sim 3\text{--}4 \cdot 10^{-3}$); singleton magnitude and partition feasibility are not in lockstep, and on Gemma the atomic positivity rate is in fact *lower* for trained (32–38 of 53) than for the scrambled

Table 1. Evidence grid. Each cell points to the table that carries the corresponding validation claim; prose elaboration follows in the subsections below.

Axis	Atomic gaps	Random feasibility	Optimized feasibility
Generalization	val drift bounded (App. G)	null rates stable on val (Table 4)	$r=10$ broad, $r=20$ selective (Table 2)
Model/protocol variation	magnitudes differ (Table 3)	Llama $\gg$ Pythia; $r=10 \gg r=20$ (Table 4)	min-gaps span $> 10\times$ (Table 3)
Trained-vs-N1 sensitivity	2.2–13 $\times$ collapse (App. G)	0/1,000 in every N1 cell (Table 7)	matched-effort GA fails (Table 3)

Table 2. Cross-corpus transfer at level (c), summary. Trained GA partition from source corpus, re-evaluated on val tensors from all three corpora. “self” = source-corpus self-eval (the partition that was optimized), averaged across the three domains for each model; “transfer” = mean min-gap across the 6 off-diagonal (source, target) pairs per model $\times r$; “#feas / 6” = number of transfers with $\min_j \Delta_j^{\Pi} \geq 0$. Per-cell breakdowns in Appendix H.

Source	r	self	mean transfer	# feas / 6
Llama	10	+0.78	+0.42	6
Llama	20	+0.37	+0.04	3
Pythia	10	+0.71	+0.51	6
Pythia	20	+0.28	+0.15	5
Gemma	10	+0.40	−0.85	1
Gemma	20	+0.06	−0.98	0

N1 control (53 of 53). At level (c), Table 3 shows trained GA-found min-gaps span -0.02 (Gemma $\times$ arxiv at $r=20$, the one infeasible trained cell) up to $+0.99$ (Llama $\times$ code at $r=10$); the GA finds feasible partitions in 17 of 18 trained cells but the margin is highly cell-dependent. This pattern establishes discriminating power, but it does not yield a clean monotonic architecture-quality ordering: model size, vocabulary size, top- K coverage, normalization, training corpus and protocol details are confounded, and Gemma $\times$ arxiv at $r=20$ shows that even a fully trained baseline can fall on the wrong side of the feasibility line.

3.3. Axis 3: Trained-Structure Sensitivity

The third validation axis asks whether log-pool feasibility depends on learned content. Under N1, tensor scales are preserved (per-tensor empirical σ_i matched) and architecture is preserved (component count N , dtype, normalization conventions identical), but learned weights are replaced by per-tensor Gaussian samples and norm gains are reset to one. For LayerNorm-folded models (Pythia), this requires an additional re-centering step on the input/output dimensions of the linear weights after sampling, to preserve the gauge under which the additive logit decomposition holds (App. A). The atomic and partition-level measurements then dissociate. At atomic resolution, the conventional sign criterion $\Delta_{\{i\}} \geq 0$ is weak: a majority of components have non-negative atomic gap in both regimes, because

Table 3. Architecture profile summary across all three measurement levels. “atomic ratio” = mean trained $\Delta_{\{i\}}$ / mean N1 $\Delta_{\{i\}}$, averaged over the three domains (per-domain values in Appendix G); “rand. feas. (uniform)” = range across the three domains of $\Pr(\min_j \Delta_j^{\Pi} \geq 0)$ on trained tensors at $n=1,000$ uniform-random partitions; “trained GA min” / “N1 GA min” = mean across the three domains of the best-found $\min_j \Delta_j^{\Pi}$ (positive = feasible), with the N1 column averaged across three init seeds.

Model	r	atomic ratio	rand. feas.	trained GA	N1 GA
Llama	10	11 $\times$	.50–.81	+0.78	−0.020
Llama	20	”	.06–.26	+0.37	−0.042
Pythia	10	4.2 $\times$	.005–.023	+0.71	−0.018
Pythia	20	”	0/1,000	+0.28	−0.058
Gemma	10	2.3 $\times$	0/1,000	+0.40	−0.420
Gemma	20	”	0/1,000	+0.06	−0.516

near-uniform $P_{\{i\}}$ at $\beta_i = 1/N$ pushes singleton gaps toward zero rather than into negative territory; the trained-vs-scrambled contrast at the atomic level therefore appears in *magnitude* (App. G), not in sign rate. At partition resolution the $\min_j \Delta_j^{\Pi} \geq 0$ criterion is an order-statistic stress test that converts the margin difference into a sharp binary separation. The optimized-partition column of Table 3 reports the trained-vs-N1 contrast for every cell: trained partitions reach -0.02 to $+0.99$ across cells (only Gemma $\times$ arxiv at $r=20$ is infeasible at the trained baseline), with ≤ 14 components in the residual group; the N1 GA cannot find any feasible partition in any of the 18 cells; the best N1 partitions leave 24 or more components in the residual group. Table 7 reports the same trained-vs-N1 contrast as a random-partition feasibility *rate*: at $n=1,000$ uniform-random partitions per cell, every Llama and Pythia cell has N1-feasibility flat at 0/1,000; every Gemma cell has zero feasibility under both regimes, but the magnitude of the negative N1 min-gap is much larger for Gemma (-1.6 to -2.9 nats) than for Llama or Pythia (-0.32 to -0.49 nats). The Gemma magnitude is consistent with Gemma’s larger logit scale (e.g. max-logit ≈ 28 in our sanity batches versus ≈ 10 for Llama and Pythia); we therefore read the cross-architecture comparison of *absolute* N1 gap magnitudes as scale-confounded, while the within-cell trained-vs-N1 sign reversal is robust.

Table 4. Trained-baseline random-partition feasibility across three null distributions, range across the three domains per cell. “0/1,000” = no feasible partition observed in 1,000 samples. Per-cell rates in Appendix E; matching N1 rates are uniformly 0/1,000 for every cell (Appendix F). *Residual convention*. All optimized-feasibility entries in Table 3 are residual-allowed: group 0 is exempt from the min-gap test, so a positive GA entry establishes feasible non-residual groupings alongside a residual, not full coverage of every architectural component. Residual-group sizes are summarized in §3 Axis 3.

Model	Setting	uniform	size	residual
Llama	$r=10$	.50–.81	.54–.93	.52–.88
Llama	$r=20$	.06–.26	.29–.56	.05–.29
Pythia	$r=10$	.005–.023	.011–.144	.000–.015
Pythia	$r=20$	0/1,000	.000–.010	.000–.001
Gemma	$r=10$	0/1,000	0/1,000	0/1,000
Gemma	$r=20$	0/1,000	0/1,000	0/1,000

4. What Is Validated?

The empirical layer of the claim is deliberately narrow: the compositional utility gap has discriminating power as a measurement tool for trained-transformer structure. It generalizes under held-out recomputation, varies across architecture and measurement settings, and in the measured cells separates trained weights from per-tensor- σ -matched scrambled weights. Importantly, the trained-vs-scrambled contrast is *resolution-dependent*: the singleton sign criterion $\Delta_{\{i\}} \geq 0$ is non-discriminative because near-uniform $P_{\{i\}}$ at $\beta_i = 1/N$ makes both regimes pass it at high rates; the trained-vs-N1 signal appears as a magnitude difference at the atomic level and as a feasibility difference at the partition level. This dissociation is itself empirical content of the metric: the same gap quantity reports different aspects of trained structure at different aggregation resolutions, and the choice of aggregation resolution determines what is being measured. Practitioners should not summarize a model by its atomic gap-sign rate; the operationally informative quantity is the partition-level $\min_j \Delta_j^{\Pi}$ together with the magnitude of the atomic distribution.

The interpretive layer remains open. The original framework (Lee et al., 2025) offers a subagent reading: feasible log-pool groups mark compositional latent agentic structure within the trained network. A more conservative distributional reading is that the gap tracks a specific property of trained output distributions (redundancy, second-order coherence, or learned-margin geometry) that happens to be detectable via this particular log-pool functional but does not imply that the constituent components are autonomous in any goal-directed sense. A third reading is abstract decomposition coherence: the metric identifies when architectural blocks admit grouped summaries whose product-of-experts composition reproduces the parent distribution under the chosen gap functional, a non-trivial but non-agentic struc-

tural fact. The paper validates the empirical content of the metric. It does not settle which of these readings is philosophically or mechanistically correct, and our results are compatible with all three.

This is where causal-abstraction and mechanistic-explanation standards enter. Faithfulness in the sense of Geiger et al. (2025) asks whether the high-level variables induced by a partition support interventionally valid counterfactuals at the low level. The compositional utility gap, as instantiated here, is a partition-existence criterion: it asks whether a candidate partition’s group distributions log-pool back into a consistent description of the parent, not whether interventions on those groups transfer. The two criteria can in principle disagree, and the natural next study is to evaluate both on the same partitions (Appendix I). Mechanistic accounts in the tradition of Craver (2007); Bechtel (2008) require decomposition, composition, organization, and activity; the gap operationalizes a necessary precondition for the decomposition step but does not establish organization or activity. Our validation makes the compositional utility gap worth using in that larger inquiry; it is not a substitute for it.

5. Scope and Controls

Five caveats. **(i) Cross-architecture is an architecture-and-protocol contrast.** Llama, Pythia, and Gemma differ in vocabulary size, top- K coverage, normalization, model size, training corpus, and logit scale; cross-architecture comparison of absolute gap magnitudes is confounded, as visible in Gemma’s $\sim 5\times$ -larger N1 random min-gaps. **(ii) Random-partition feasibility is architecture-specific.** Only Llama is permissive at $r=10$; Pythia is rare and Gemma zero. The size-matched-null bonus over uniform-random (Table 4) shows that partition *geometry* alone explains part of the trained-baseline feasibility, before learned weight structure is invoked. **(iii) N1 admits multiple collapse mechanisms** (absent trained structure, near-uniform scrambled outputs, or magnitude/temperature effects), disambiguated in Appendix I. **(iv) Optimized-level results are GA-best-found**, not worst-case or expected; cross-seed reproducibility is verified at the random-partition level (Appendix E). **(v) Singleton sign $\Delta_{\{i\}} \geq 0$** is high in both regimes (in fact higher under N1, e.g. 53/53 for Gemma N1 vs 32–38/53 trained); this is an artifact of $\beta_i = 1/N$ pushing $P_{\{i\}}$ toward uniform. Atomic discriminates by *magnitude*, not sign; partition-level $\min_j \Delta_j^{\Pi}$ is the informative resolution.

References

Bechtel, W. *Mental Mechanisms: Philosophical Perspectives on Cognitive Neuroscience*. Routledge, New York, 2008.

- Biderman, S., Schoelkopf, H., Anthony, Q., Bradley, H., O’Brien, K., Hallahan, E., Khan, M. A., Purohit, S., Prashanth, U. S., Raff, E., et al. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 2397–2430. PMLR, 2023.
- Craver, C. F. *Explaining the Brain: Mechanisms and the Mosaic Unity of Neuroscience*. Oxford University Press, Oxford, 2007.
- Geiger, A., Ibeling, D., Zur, A., Chaudhary, M., Chauhan, S., Huang, J., Arora, A., Wu, Z., Goodman, N. D., Potts, C., and Icard, T. Causal abstraction: A theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research*, 26(83):1–64, 2025. URL <https://www.jmlr.org/papers/v26/23-0058.html>.
- Gemma Team, Riviere, M., Pathak, S., Sessa, P. G., et al. Gemma 2: Improving open language models at a practical size, 2024.
- Grattafiori, A., Dubey, A., Jauhri, A., et al. The Llama 3 herd of models, 2024.
- Grote, T., Genin, K., and Sullivan, E. Reliability in machine learning. *Philosophy Compass*, 19(5):e12974, 2024. doi: 10.1111/phc3.12974.
- Hubinger, E., van Merwijk, C., Mikulik, V., Skalse, J., and Garrabrant, S. Risks from learned optimization in advanced machine learning systems, 2019.
- Lee, S. H., Kondor, R., and Ngo, R. Probabilistic modeling of latent agentic substructures in deep neural networks, 2025. URL <https://arxiv.org/abs/2509.06701>.

A. Architectures and Logit Decomposition

Three architectures. The three transformers in the empirical inventory are summarised below; all have bias-free attention and bias-free unembedding ($b_O = b_Q = b_K = b_V = b_U = 0$).

- **Llama 3.1 8B** (Grattafiori et al., 2024): 32 layers, $d_{\text{model}} = 4,096$, 32 attention heads (grouped-query: 8 KV heads), vocabulary 128,256, RMSNorm pre-attention and pre-MLP only (no post-norms). No final-logit soft-capping.
- **Pythia 1.4B** (Biderman et al., 2023): 24 layers, $d_{\text{model}} = 2,048$, 16 multi-head attention heads, vocabulary 50,304, LayerNorm pre-attention and pre-MLP. No final-logit soft-capping.
- **Gemma 2 2B** (Gemma Team et al., 2024): 26 layers, $d_{\text{model}} = 2,304$, 8 attention heads (grouped-query: 4 KV heads), vocabulary 256,000, RMSNorm *both* pre- and post-attention/MLP, and a final-logit soft-cap $z' = 30 \cdot \tanh(z/30)$.

Under the layer-level decomposition each model contributes $N = 1 + 2L$ components (embedding, L attention sublayers, L MLP sublayers): Llama $N = 65$, Pythia $N = 49$, Gemma $N = 53$.

Residual-stream additivity. The final residual stream is the sum of all component contributions:

$$h^{(L)}(x) = C_{\text{embed}}(x) + \sum_{\ell=1}^L (C_{\text{attn},\ell}(x) + C_{\text{mlp},\ell}(x)).$$

Intermediate per-layer pre-norms (RMSNorm/LayerNorm applied inside each sublayer) do not break this additivity at the residual-stream level; they affect only how a downstream component’s output is computed from the (additive) input it receives.

Final normalization and the RMS factor. The final logits are $z(x) = \text{Norm}(h^{(L)}(x)) \cdot W_U$, where Norm is RMSNorm (Llama, Gemma) or LayerNorm (Pythia). For RMSNorm, $\text{Norm}(h) = \frac{\gamma \odot h}{\text{RMS}(h)}$ with $\text{RMS}(h) = \sqrt{\frac{1}{d} \sum_i h_i^2 + \varepsilon}$; the per-input scalar $1/\text{RMS}(h^{(L)})$ does *not* cancel in the softmax. We therefore record $\text{RMS}(h^{(L)})$ once per input and apply it linearly to each component’s contribution:

$$z_i(x) = \frac{1}{\text{RMS}(h^{(L)}(x))} \cdot W_U \cdot \text{diag}(\gamma) \cdot C_i(x).$$

Because RMSNorm is element-wise and γ is fixed, $\sum_i z_i = z$ holds exactly. For LayerNorm (Pythia), centering and scaling factors are likewise per-input scalars and the analogous identity holds. In TransformerLens with the *fold-LN* convention (its default; we use it), γ is folded into W_U at load time, so the working formula is $z_i = (C_i/\text{RMS}(h^{(L)})) \cdot W_U$.

Soft-capping (Gemma 2 only). Gemma 2 applies $z'(x) = 30 \cdot \tanh(z(x)/30)$ as a final transformation. Because $\tanh$ is non-linear, $\sum_i \tanh(z_i) \neq \tanh(\sum_i z_i)$, and exact additive decomposition fails on capped logits. We use *pre-capped* logits throughout. The HuggingFace Gemma 2 release notes report “very minor” downstream effects when soft-capping is disabled, and Gemma 3 removed it entirely (replaced with QK-norm); the pre-cap convention is also the default in TransformerLens with the soft-cap configuration disabled and in Flash Attention implementations.

Verification. We assert the additive identity per-input on each new top- K tensor at extraction time: $\|\sum_i z_i - z\|_\infty < 10^{-3}$ at fp32 or $\|\sum_i z_i - z\|_\infty / \max |z| < 5\%$ at bf16 (the realistic floor given bf16’s ~ 3 -decimal mantissa precision). The threshold is dtype-aware and hard-fails the pipeline; a tensor that breaks the identity is never propagated to downstream statistics.

B. The Log-Pool Compositional Framework

Fix a finite outcome space $\mathcal{Y}$, such as a tokenizer vocabulary, and a context x . All distributions below are conditional on x and are assumed strictly positive on $\mathcal{Y}$, as is automatic for softmax distributions. We write the conditioning explicitly when needed. The framework of Lee et al. (2025) starts from component distributions $P_1, \dots, P_n$ and non-negative weights $\beta_1, \dots, \beta_n$ satisfying $\sum_i \beta_i = 1$. Their weighted logarithmic pool is

$$P_\beta(y | x) = \frac{\exp(\sum_{i=1}^n \beta_i \log P_i(y | x))}{\sum_{y' \in \mathcal{Y}} \exp(\sum_{i=1}^n \beta_i \log P_i(y' | x))} = \frac{\prod_{i=1}^n P_i(y | x)^{\beta_i}}{\sum_{y' \in \mathcal{Y}} \prod_{i=1}^n P_i(y' | x)^{\beta_i}}. \quad (1)$$

Thus the aggregate distribution is a normalized geometric mean of the component distributions. Equivalently, the component log-probabilities are averaged and then renormalized. This is not the arithmetic mean $\sum_i \beta_i P_i(y | x)$, which is the linear opinion pool and is not the composition rule used in this paper.

The welfare motivation is log-score based. A component with belief distribution Q is assigned epistemic utility

$$W_Q(y | x) = \log Q(y | x). \quad (2)$$

If a larger composition has distribution P , then the component’s welfare inside the composition is evaluated by $\mathbb{E}_{y \sim P(\cdot | x)}[\log Q(y | x)]$, whereas its solo welfare is $\mathbb{E}_{y \sim Q(\cdot | x)}[\log Q(y | x)]$. The cited framework calls the component compositional when the former is at least the latter. We use this motivation to state the quantity transparently; our empirical interpretation below does not require treating log-score welfare as the only possible reading of the metric.

For a candidate component or subset S , let $P_S(\cdot | x)$ denote the distribution assigned to that subset by the chosen decomposition. In the abstract log-pool setting, P_S may itself be the logarithmic pool of the members of S . In our transformer experiments, P_S is the softmax distribution induced by the subset’s additive logit contribution, with the precise temperature convention fixed by the extraction protocol. Given a prompt distribution or evaluation dataset D , the compositional utility gap is

$$\Delta_S(P) := \mathbb{E}_{x \sim D} \left[\sum_{y \in \mathcal{Y}} (P(y | x) - P_S(y | x)) \log P_S(y | x) \right]. \quad (3)$$

Equivalently,

$$\Delta_S(P) = \mathbb{E}_{x \sim D} \left[\sum_y P(y | x) \log P_S(y | x) - \sum_y P_S(y | x) \log P_S(y | x) \right] \quad (4)$$

$$= \mathbb{E}_{x \sim D} [-\text{CE}(P(\cdot | x), P_S(\cdot | x)) + H(P_S(\cdot | x))] \quad (5)$$

$$= \mathbb{E}_{x \sim D} [H(P_S(\cdot | x)) - H(P(\cdot | x)) - \text{KL}(P(\cdot | x) \| P_S(\cdot | x))]. \quad (6)$$

The last equality uses $\text{CE}(P, P_S) = H(P) + \text{KL}(P \| P_S)$. For a fixed parent distribution P and fixed dataset D , the term $\mathbb{E}_{x \sim D} H(P(\cdot | x))$ is constant across candidate subsets. Pointwise in x , the non-negativity condition is therefore

$$\text{KL}(P(\cdot | x) \| P_S(\cdot | x)) \leq H(P_S(\cdot | x)) - H(P(\cdot | x)), \quad (7)$$

and after averaging over prompts,

$$\Delta_S(P) \geq 0 \iff \mathbb{E}_{x \sim D} \text{KL}(P(\cdot | x) \| P_S(\cdot | x)) \leq \mathbb{E}_{x \sim D} [H(P_S(\cdot | x)) - H(P(\cdot | x))]. \quad (8)$$

This form is useful because it shows that the gap is a relation between two distributions. A positive gap requires P_S to be sufficiently close to the parent distribution P , as measured by forward KL, and sufficiently high entropy relative to P to pay for that mismatch.

Operationally, a “compositional agent” in this framework is a distribution whose own log-score is no worse under the composed distribution than under itself. For our subset notation this means

$$\Delta_S(P) \geq 0 \iff \mathbb{E}_{x, y \sim P} \log P_S(y | x) \geq \mathbb{E}_{x \sim D, y \sim P_S(\cdot | x)} \log P_S(y | x), \quad (9)$$

where both sides use the same prompt distribution. Intuitively, the parent distribution assigns at least as much average log-score, according to P_S ’s own scoring rule, as P_S assigns to itself. A family of components is unanimously compositional when this non-negativity condition holds for every member or group under consideration.

The information-theoretic identity also marks the boundary of the formal claim. Non-negativity of Δ_S is not an architectural property by itself. It does not say that S is a causal module, that it is independently executable, that interventions on S transfer to the full model, or that S is agentic in any goal-directed sense. It says that, under the chosen distributions and chosen log-pool weighting convention, the parent distribution and the subset distribution stand in the entropy–KL relation in Eq. (6). The framework of [Lee et al. \(2025\)](#) does not supply a theorem licensing the inference from $\Delta_S \geq 0$ to “this subset

of a trained network is a mechanistic subagent.” Its formal move is conservative: it defines a welfare criterion on probability distributions and studies when log-pooled distributions satisfy it.

Our paper adds a separate empirical identification step. For each trained transformer and prompt x , we decompose the final next-token logits additively:

$$z(x) = \sum_{i \in \mathcal{I}} z_i(x), \quad P(\cdot | x) = \text{softmax}(z(x)). \quad (10)$$

The index set $\mathcal{I}$ contains architectural blocks such as the embedding contribution, per-layer attention sublayers, and per-layer MLP sublayers. For a group $S \subseteq \mathcal{I}$, we form

$$z_S(x) = \sum_{i \in S} z_i(x). \quad (11)$$

In the mechanistic product-of-experts convention, the group distribution is $P_S(\cdot | x) = \text{softmax}(z_S(x))$. In the normalized weighted convention, we choose weights $\beta_i > 0$ with $\sum_i \beta_i = 1$, set $\beta_S = \sum_{i \in S} \beta_i$, and use the temperature-adjusted distribution

$$\tilde{P}_S(\cdot | x) = \text{softmax}(z_S(x)/\beta_S). \quad (12)$$

With this convention, the full model distribution is exactly recovered as the weighted log-pool of the temperature-adjusted component distributions:

$$P(y | x) \propto \prod_{i \in \mathcal{I}} \tilde{P}_i(y | x)^{\beta_i}. \quad (13)$$

This establishes the algebraic bridge from residual-stream logit additivity to the log-pool rule. It does not by itself establish that the blocks are the unique, privileged, or causally autonomous parts of the model.

The resulting empirical question is therefore deliberately narrower than a general part-detection claim. We ask whether architecturally specified decompositions of trained transformers produce subset distributions with $\Delta_S \geq 0$, how often this occurs, how it varies across layers and block types, and whether candidate partitions can make all groups non-negative simultaneously. Positivity of the gap is a necessary condition for the subset to count as compositional under this log-score/log-pool criterion, but it is not sufficient for substantive part-attribution. Any parent distribution can be paired with many possible decompositions and many admissible β weightings, and changing that identification can change the measured gaps. The empirical contribution of this paper is to test the discriminating power and stability of the gap as a measurement tool for concrete transformer decompositions, not to treat the gap as an automatic detector of mechanistic agency.

C. Methodological Details

Top- K truncation and validation tensors. For every trained (model, domain) cell we extract a held-out validation tensor of shape $[M, K, N]$ with $M = 1,000$ chunks of length 128, $K = 10,000$ top-vocabulary tokens selected per chunk by that model’s own parent logits, and N components per the layer-level decomposition above. For N1, the same text chunks and extraction protocol are reused, but the top- K support is selected from the N1 model’s own parent logits, not copied from the trained model’s top- K token set; trained-vs-N1 comparisons therefore differ in both the per-chunk top- K subset and the per-component logits, never in the input text. For the optimized-partition genetic algorithm we additionally extract train tensors at $M = 3,000$ chunks from the trained-baseline pipeline; the same train tokens are reused for the matching N1 train extraction.

Top- K coverage asymmetry. Because vocabulary sizes differ across architectures, the fraction of the vocabulary captured at $K = 10,000$ also differs: Llama retains 7.8% of 128K tokens, Pythia retains 19.9% of 50K, Gemma retains 3.9% of 256K. Cross-architecture magnitude comparisons of the gap should be read with this asymmetry in mind; the within-architecture trained-vs-N1 contrast is the cleaner inference.

Mixing weights. The log-pool of Lee et al. (2025) permits any non-negative simplex weighting β_i . The atomic singleton diagnostic uses the uniform choice $\beta_i = 1/N$, producing $P_{\{i\}} = \text{softmax}(z_i/N)$; the partition experiments use the unnormalized product-of-experts convention introduced in §2 (equivalent to $\beta_i = 1$ within each group). Both choices are summed-additive over the architectural decomposition; neither couples to a learned β . Re-estimating partition results under the exact simplex-normalized reconstruction $P_S = \text{softmax}(z_S/\beta_S)$ of Lee et al. (2025) (see Appendix B) is left for follow-up.

Compositionality gap on partitions. A partition is $\Pi : \{1, \dots, N\} \rightarrow \{0, 1, \dots, r\}$; group 0 is residual (excluded from the feasibility test) and non-empty groups $j \in \{1, \dots, r\}$ are evaluated by $\Delta_j^\Pi = \mathbb{E}_x[\sum_y (P(y|x) - P_j^\Pi(y|x)) \log P_j^\Pi(y|x)]$. For partition experiments we use the stated product-of-experts group convention,

$$l_j^\Pi = \sum_{i:\Pi(i)=j} z_i, \quad P_j^\Pi = \text{softmax}(l_j^\Pi),$$

over the top- K tokens; this is the same unnormalized convention introduced in §2. A partition is *feasible* when $\min_j \Delta_j^\Pi \geq 0$ over its non-empty non-residual groups. Because the residual group is exempt from the feasibility test, a feasible partition only shows that the assigned groups satisfy the criterion conditional on the residual; it does not show that every architectural component participates in a fully covered compositional partition. We track the residual count alongside feasibility for each reported partition.

Genetic algorithm and random-null evaluation. Optimized partitions are produced by the genetic algorithm specified in Appendix D, run identically on trained-baseline and N1 tensors. The search fits on the train tensor; the saved best partition is then recomputed on the held-out val tensor (this val recomputation is what the paper reports throughout). Random-partition null sampling at $n = 1,000$ samples per cell uses the same val tensor.

Parameter-scrambled (N1) control. The N1 reinit replaces every weight tensor with $\mathcal{N}(0, \hat{\sigma}_i^2)$ where $\hat{\sigma}_i$ is the trained tensor’s empirical standard deviation; norm gains are reset to 1 and all biases to 0. This preserves architecture, dtype convention, per-tensor scale, and the extraction code path while destroying learned correlations. The N1 reinit is deterministic in the init seed; we use seeds {42, 43, 44} throughout to estimate seed variance.

Per-cell dtype matching. Each N1 cell uses the same dtype as the corresponding trained-baseline extraction: Llama-WikiText fp32, Llama-arxiv bf16, Llama-code bf16, Pythia all-domains fp32, Gemma all-domains fp32. This ensures that trained-vs-N1 contrasts are not confounded by precision differences.

D. Genetic Algorithm for Partition Search

The optimized-partition column of Table 3 reports the result of a single search procedure applied uniformly to every trained-baseline and N1 cell. This appendix specifies the procedure.

Search space. An assignment is a vector $\Pi \in \{0, 1, \dots, r\}^N$ that labels each architectural component with a group index. Index 0 is the residual; indices $1, \dots, r$ are the assigned groups. The raw assignment space has $(r+1)^N$ elements, which is the reason we do not enumerate. In the CMAD mode used throughout the paper, empty assigned groups are permitted (so larger r subsumes search at smaller $r' < r$); only the all-residual case is repaired, by moving one component from the residual into group 1 to keep fitness well-defined.

Fitness. The genetic algorithm maximizes

$$\text{fit}(\Pi) = \min_j \Delta_j^\Pi + \mu \cdot \frac{N - n_0(\Pi)}{N},$$

where the first term is the worst non-residual group gap (the quantity used for feasibility; §2) and the second is a coverage bonus that scales linearly with the fraction of components assigned to a non-residual group. Here $n_0(\Pi)$ is the residual count and $\mu = 1.0$ is fixed across cells. Feasibility is determined only by $\min_j \Delta_j^\Pi \geq 0$ over non-empty non-residual groups; the coverage term is a search regularizer, not part of the feasibility criterion. Every table in the paper reports the validation $\min_j \Delta_j^\Pi$ separately from GA fitness.

Selection. Each generation runs tournament selection with tournament size 5: five individuals are drawn uniformly at random with replacement; the highest-fitness member is chosen as a parent. Two parent draws produce one offspring pair.

Crossover (rate 0.7). Group labels of one parent are permuted to maximize overlap with the other parent (a Hungarian alignment over the $r \times r$ overlap matrix), so that label permutations of the same partition produce the same child. After

alignment, each component independently inherits its label from one of the two parents with equal probability. Residual-group assignments pass through directly and are not part of the alignment. If the offspring has no non-empty non-residual group, one component is moved out of the residual into group 1 to keep fitness well-defined; empty assigned groups beyond that are permitted.

Mutation. Two mutation operators are applied to each offspring. The first reassigns each component independently with probability 0.10 to a uniform-random different group. The second applies $\max(1, \lfloor 0.03 N \rfloor)$ attempted pair swaps; each attempt picks two random component indices and swaps their group labels if they differ. With the component counts used here ($N \in \{49, 53, 65\}$), this is one attempted swap per offspring. The repair step from the crossover paragraph is applied again after mutation.

Elitism and population update. The top 10% of the current population by fitness is copied unchanged into the next generation; the remaining 90% is generated by the selection + crossover + mutation pipeline. The population size is 500 throughout. The run terminates after 500 generations.

Evaluation. Per-generation fitness is first screened on a deterministic proxy consisting of the first $\min(300, M) = 300$ chunks of the train tensor (a fixed subset, not resampled across generations). The top $\max(10, \lfloor 0.4 \cdot \text{pop} \rfloor) = 200$ proxy-ranked offspring are then re-evaluated on all $M = 3,000$ train chunks; the remaining offspring keep their proxy scores when the population is ranked. The parameter `batch_size = 256` is the GPU mini-batch size used inside each evaluation pass; it is a memory-management knob, not the proxy size. After the GA terminates, the best-fitness individual tracked across the entire run (best-ever tracking, not the final generation alone) is recomputed on the held-out val tensor at $M = 1,000$; the resulting $\text{val } \min_j \Delta_j^\Pi$ is what every table in the paper reports.

Initialization and determinism. The initial 500-individual population combines deterministic seeds (baseline partitions, plus depth-contiguous block partitions with three small random perturbations) with uniform-random assignments that fill the remainder of the pool. The random seed is fixed at 42 for all reported cells, giving deterministic search trajectories given the input tensor.

What a single run returns. Each saved partition is the best-fitness individual tracked across all generations of the GA run, ranked by the mixed full/proxy evaluation described above. We do not run the search multiple times per cell; cross-seed reproducibility of the feasibility criterion is checked at the random-partition level (Appendix E) rather than at the GA level. This is the basis for the caveat (§5, item (iv)) that optimized-level results report what a single search found, not what is typical of all feasible solutions.

E. Trained-Baseline Random-Partition Feasibility

F. N1 Parameter-Scrambled Results

Table 6 reports the per-cell uniform-random null feasibility rates for every $(\text{model}, \text{domain}, r, \text{seed})$ tuple in the N1 inventory.

The pattern is uniform across the full 54-cell inventory ($3 \text{ models} \times 3 \text{ domains} \times 2 \text{ resolutions} \times 3 \text{ init seeds}$): every random partition out of 1,000 samples per cell is infeasible. The p99 column (best-of-1,000) sits at -0.26 to -0.35 nats for Llama and Pythia and at -1.59 to -2.22 nats for Gemma: deeply infeasible by orders of magnitude in either case. For the matching trained-baseline numbers, see Appendix E; the trained-vs-N1 contrast is largest at Llama-arxiv $r=10$ ($0.808 \rightarrow 0$) and is trivially small at the trained cells that already had zero random feasibility (Pythia $r=20$ across domains; all Gemma cells), which serve as consistency checks rather than as new evidence.

Table 7 consolidates the same contrast at the $(\text{model}, \text{domain}, r)$ granularity for direct trained-vs-N1 inspection.

G. Atomic Gap Distributions

Per-component atomic gaps $\Delta_{\{i\}} = \mathbb{E}_x[\sum_y (P - P_{\{i\}}) \log P_{\{i\}}]$ with $P_{\{i\}} = \text{softmax}(z_i/N)$ and $\beta_i = 1/N$. Top- $K = 10,000$ vocabulary truncation; $M = 1,000$ held-out validation chunks per cell.

Table 5. Trained-baseline random-partition feasibility rates ($n = 1,000$ partitions) for each cell against three null distributions (uniform-random, size-matched, residual-matched). All rates are recomputed on validation tensors.

Model	domain	r	uniform	size-matched	residual-matched
Llama	WikiText	10	0.499	0.536	0.519
Llama	arXiv	10	0.808	0.912	0.877
Llama	code	10	0.722	0.932	0.782
Llama	WikiText	20	0.065	0.293	0.053
Llama	arXiv	20	0.257	0.562	0.293
Llama	code	20	0.064	0.363	0.069
Pythia	WikiText	10	0.005	0.011	0.000
Pythia	code	10	0.021	0.086	0.011
Pythia	arXiv	10	0.023	0.144	0.015
Pythia	WikiText	20	0.000	0.000	0.000
Pythia	code	20	0.000	0.010	0.001
Pythia	arXiv	20	0.000	0.008	0.000
Gemma	WikiText	10	0.000	0.000	0.000
Gemma	arXiv	10	0.000	0.000	0.000
Gemma	code	10	0.000	0.000	0.000
Gemma	WikiText	20	0.000	0.000	0.000
Gemma	arXiv	20	0.000	0.000	0.000
Gemma	code	20	0.000	0.000	0.000

Sign vs. magnitude. The $\Delta_{\{i\}} \geq 0$ count is high in both regimes, in fact *slightly higher under N1*: trained Gemma is 32–38 of 53 (60–72%), trained Llama 54–59 of 65 (83–91%), trained Pythia 43–44 of 49 (88–90%); N1 averages are uniformly above 89% for Llama and Pythia, and exactly 100% for Gemma in every domain. Sign rate is therefore not a discriminating statistic. What does discriminate is the *magnitude* of the gap: trained means lie in the $3\text{--}11 \times 10^{-3}$ range while N1 means lie in the 3×10^{-4} (Llama), 1×10^{-3} (Pythia) and 4×10^{-3} (Gemma) ranges. The trained-vs-N1 ratio in mean magnitude is $\sim 10\text{--}13\times$ for Llama, $\sim 3.5\text{--}5.4\times$ for Pythia, and only $\sim 2.2\text{--}2.6\times$ for Gemma: a clear architecture-dependent ranking.

Interpretation. At $\beta_i = 1/N$, singleton logits $\beta_i z_i$ are weak; the induced $P_{\{i\}}$ sits close to uniform. The gap $\Delta_{\{i\}}$ is then small in absolute terms regardless of whether weights are trained. Near-uniform $P_{\{i\}}$ also makes the second-order term in the gap collapse toward zero rather than swinging negative, which is why even N1 components mostly clear the ≥ 0 threshold. Training does not lift gaps from negative to positive; it lifts them from near-zero positive to non-trivially positive. This rescaling is what level (b) (random-partition feasibility) and level (c) (search-found feasibility) exploit, because $\min_j \Delta_j^\Pi$ is an order statistic over groups: a partition is feasible only when every group’s gap clears zero, and trained models supply enough margin for many random groupings to clear while N1 groupings hover around zero and routinely fall below.

H. Cross-Corpus Transfer

For each trained-baseline GA-found partition, Table 9 reports the recomputed $\min_j \Delta_j^\Pi$ both on its source corpus (self) and on the other two corpora (transfer).

At $r = 10$, every Llama and Pythia self-eval cell transfers feasibly to every other corpus of the same model (six Llama transfers and six Pythia transfers, all positive); Pythia $\times$ arxiv at $r = 10$ has the single largest off-diagonal min-gap in the table (+1.14 on code). At $r = 20$, Llama and Pythia transfer becomes selective: Llama $\times$ code and Llama $\times$ arxiv partitions fail self-eval feasibility on WikiText, and Pythia $\times$ code fails on WikiText. The Gemma rows tell a different story: only one of twelve Gemma transfer cells is feasible (Gemma $\times$ arxiv $r=10$ to WikiText, +0.025), and Gemma $\times$ code partitions transfer catastrophically ($\min_j \Delta_j^\Pi$ around -2.0 on WikiText and arxiv). Read these numbers in light of Appendix E: Llama $\times$ {arxiv, code} at $r = 10$ have trained-baseline random-partition feasibility above 73%, so “transfer succeeded” there is partly explained by the criterion being permissive on the target corpus anyway. The non-trivial transfer signal lives at $r = 20$ and on the Gemma cells, where the random feasibility is at or near zero even under the size-matched null.

I. Future Disambiguating Controls

The trained-vs-N1 contrast at level (b) is robust (uniformly 0/1,000 random feasibility across all 54 N1 seed-cells against trained-baseline rates that range from 0/1,000 to 0.93 depending on (model, domain, r)), but it does not, on its own,

Table 6. Uniform-random N1 null feasibility ($\min_j \Delta_j^\Pi \geq 0$, the paper-wide feasibility threshold) at $n_{\text{null}} = 1,000$ random partitions per cell. “mean” is the mean of the per-partition $\min_j \Delta_j^\Pi$ across the 1,000 samples; “p99” is the 99th percentile (the top-of-distribution random partition in the cell). Every $\Pr(\text{feasible}) = 0/1,000$; the epsilon-band column $\Pr(\min_j \Delta_j^\Pi \geq -10^{-3})$ is also 0 in every measured cell, confirming the collapse is not a boundary effect.

Model	Domain	Seed	r	$\Pr(\text{feas})$	mean $\min_j \Delta_j^\Pi$	p99 $\min_j \Delta_j^\Pi$
Llama	WikiText	42	10	0.000	−0.500	−0.359
Llama	WikiText	42	20	0.000	−0.441	−0.326
Llama	WikiText	43	10	0.000	−0.503	−0.365
Llama	WikiText	43	20	0.000	−0.434	−0.324
Llama	WikiText	44	10	0.000	−0.471	−0.338
Llama	WikiText	44	20	0.000	−0.401	−0.304
Llama	code	42	10	0.000	−0.478	−0.341
Llama	code	42	20	0.000	−0.416	−0.325
Llama	code	43	10	0.000	−0.485	−0.354
Llama	code	43	20	0.000	−0.420	−0.313
Llama	code	44	10	0.000	−0.477	−0.342
Llama	code	44	20	0.000	−0.408	−0.314
Pythia	WikiText	42	10	0.000	−0.461	−0.332
Pythia	WikiText	42	20	0.000	−0.382	−0.261
Pythia	WikiText	43	10	0.000	−0.477	−0.347
Pythia	WikiText	43	20	0.000	−0.399	−0.272
Pythia	WikiText	44	10	0.000	−0.477	−0.346
Pythia	WikiText	44	20	0.000	−0.398	−0.277
Pythia	code	42	10	0.000	−0.474	−0.332
Pythia	code	42	20	0.000	−0.393	−0.268
Pythia	code	43	10	0.000	−0.491	−0.356
Pythia	code	43	20	0.000	−0.410	−0.284
Pythia	code	44	10	0.000	−0.470	−0.346
Pythia	code	44	20	0.000	−0.394	−0.276

discriminate between three live mechanisms for why N1 collapses: (a) the trained residual/logit structure admits log-pool feasibility while random structure does not; (b) random-init outputs sit close to uniform, making P_S also close to uniform and pushing Δ_S toward zero by a softmax-collapse mechanism orthogonal to learning; (c) the trained logit magnitudes simply scale the gap into an informative numerical range while N1 logit magnitudes do not.

Five interventional controls would discriminate these mechanisms and tighten the inference structure.

Temperature-scaling N1 logits to match trained magnitudes. Sample-by-sample, scale the N1 model’s logits by a constant τ chosen so that the empirical std of z_{N1} matches the trained-model std. Recompute random-partition feasibility. If feasibility recovers under temperature-matching, mechanism (c) (magnitude effect) is doing the work and the trained-vs-N1 contrast is partly an artifact of trained logits being scaled into a more discriminative range. If feasibility stays near zero after temperature-matching, magnitude alone does not explain the gap.

Temperature-scaling trained logits down to N1 magnitude. The complementary direction: scale trained logits so their std matches N1, then check whether random-partition feasibility on trained collapses to N1 rates. If trained feasibility survives this contraction, learned structure is doing more than scaling and (a) is supported. If it collapses, scaling carries most of the signal.

Shuffled-trained controls. Permute trained weight tensor entries within each tensor. This preserves the empirical std (matched per-tensor) and the marginal distribution of per-component contributions, but destroys component identity and any cross-tensor correlation. Compare random-partition feasibility on shuffled-trained against both trained and N1. A shuffled-trained that matches N1 indicates the trained-vs-N1 gap is about cross-component correlation; one that matches trained indicates the gap is about per-tensor distributional shape rather than identity-specific structure.

Table 7. Trained-vs-N1 random-partition feasibility contrast at level (b). $\Pr(\text{feasible})$ at $n=1,000$ uniform-random partitions per cell; N1 averaged over three parameter-scrambling seeds. “mean” and “p99” are the across-partition mean and 99th percentile (top 10 of 1,000) of $\min_j \Delta_j^\Pi$ on N1 tensors.

Model	Domain, r	Pr(feasible)		$\min_j \Delta_j^\Pi$ on N1 (nats)	
		Trained	N1	mean	p99
Llama	WikiText, $r=10$	0.499	0/1,000	-0.491	-0.354
Llama	WikiText, $r=20$	0.065	0/1,000	-0.425	-0.318
Llama	arxiv, $r=10$	0.808	0/1,000	-0.491	-0.353
Llama	arxiv, $r=20$	0.257	0/1,000	-0.425	-0.324
Llama	code, $r=10$	0.722	0/1,000	-0.480	-0.345
Llama	code, $r=20$	0.064	0/1,000	-0.415	-0.317
Pythia	WikiText, $r=10$	0.005	0/1,000	-0.471	-0.345
Pythia	WikiText, $r=20$	0.000	0/1,000	-0.395	-0.278
Pythia	arxiv, $r=10$	0.023	0/1,000	-0.442	-0.324
Pythia	arxiv, $r=20$	0.000	0/1,000	-0.370	-0.261
Pythia	code, $r=10$	0.021	0/1,000	-0.442	-0.325
Pythia	code, $r=20$	0.000	0/1,000	-0.370	-0.261
Gemma	WikiText, $r=10$	0.000	0/1,000	-2.940	-2.223
Gemma	WikiText, $r=20$	0.000	0/1,000	-2.284	-1.606
Gemma	arxiv, $r=10$	0.000	0/1,000	-2.937	-2.215
Gemma	arxiv, $r=20$	0.000	0/1,000	-2.281	-1.587
Gemma	code, $r=10$	0.000	0/1,000	-2.938	-2.208
Gemma	code, $r=20$	0.000	0/1,000	-2.282	-1.588

Table 8. Atomic gap summary. “ n_+ ” is the count of components with $\Delta_{\{i\}} \geq 0$ out of N ; “mean” is the across-component mean of $\Delta_{\{i\}}$; “max” is the across-component maximum. N1 figures are averaged across three init seeds {42, 43, 44}, across the three domains {WikiText, arxiv, code} where reported per-architecture in Table 3.

Model	Domain	Trained			N1 (mean over 3 seeds)		
		n_+/N	mean	max	n_+/N	mean	max
Llama	WikiText	57/65	$3.2 \cdot 10^{-3}$	$4.1 \cdot 10^{-2}$	61/65	$3.0 \cdot 10^{-4}$	$2.9 \cdot 10^{-3}$
Llama	arxiv	59/65	$3.2 \cdot 10^{-3}$	$4.1 \cdot 10^{-2}$	64/65	$3.0 \cdot 10^{-4}$	$2.6 \cdot 10^{-3}$
Llama	code	54/65	$4.0 \cdot 10^{-3}$	$6.3 \cdot 10^{-2}$	64/65	$3.0 \cdot 10^{-4}$	$2.6 \cdot 10^{-3}$
Pythia	WikiText	44/49	$3.9 \cdot 10^{-3}$	$1.2 \cdot 10^{-2}$	48/49	$1.0 \cdot 10^{-3}$	$4.6 \cdot 10^{-3}$
Pythia	arxiv	43/49	$3.8 \cdot 10^{-3}$	$1.3 \cdot 10^{-2}$	48/49	$1.1 \cdot 10^{-3}$	$4.8 \cdot 10^{-3}$
Pythia	code	44/49	$5.7 \cdot 10^{-3}$	$2.5 \cdot 10^{-2}$	48/49	$1.1 \cdot 10^{-3}$	$4.6 \cdot 10^{-3}$
Gemma	WikiText	38/53	$1.1 \cdot 10^{-2}$	$1.6 \cdot 10^{-1}$	53/53	$4.0 \cdot 10^{-3}$	$1.4 \cdot 10^{-2}$
Gemma	arxiv	32/53	$1.0 \cdot 10^{-2}$	$1.9 \cdot 10^{-1}$	53/53	$4.0 \cdot 10^{-3}$	$1.4 \cdot 10^{-2}$
Gemma	code	35/53	$1.0 \cdot 10^{-2}$	$1.2 \cdot 10^{-1}$	53/53	$4.0 \cdot 10^{-3}$	$1.4 \cdot 10^{-2}$

Architecture-respecting nulls. Our current nulls (uniform-random, size-matched, residual-matched) are exchange-symmetric over the component index. Architecture-respecting nulls (distance-matched: partitions whose groups respect graph distance in the residual stream; depth-matched: groups span similar depth bands; type-matched: groups respect the embed/attn/MLP distinction) would tighten the licensing claim from “non-generic vs exchange-symmetric random” to “non-generic vs architecturally plausible random.”

Joint gap-and-faithfulness evaluation. On the same set of GA-found partitions, evaluate both $\min_j \Delta_j^\Pi \geq 0$ feasibility and Geiger-style (Geiger et al., 2025) causal-abstraction faithfulness (e.g., interchange intervention accuracy against a high-level causal model induced by the partition). Convergence of the two licensing tests strengthens the case for the gap result tracking interventionally meaningful structure; divergence places the metric in a different category and clarifies what it actually measures.

These controls are out of scope for the workshop submission. Item 1 (temperature-scaling N1 up to trained magnitude) is the cheapest and most diagnostic, and is the natural next experiment.

Table 9. Cross-corpus transfer of trained GA partitions. Each row is a saved partition; columns are the recomputed min-gap on val tensors from each target corpus. Bold = feasible (positive); shading indicates the self-eval column. Read in light of Appendix E: high transfer is partly explained by the source corpus’s already-permissive random-feasibility rate (Llama at $r = 10$).

Source partition	r	Target ($\min_j \Delta_j^\Pi$ on val)		
		WikiText	arxiv	code
Llama $\times$ WikiText	10	0.657 (self)	0.572	0.598
Llama $\times$ WikiText	20	0.312 (self)	0.279	0.239
Llama $\times$ arxiv	10	0.303	0.681 (self)	0.666
Llama $\times$ arxiv	20	−0.082	0.289 (self)	0.283
Llama $\times$ code	10	0.041	0.366	0.993 (self)
Llama $\times$ code	20	−0.446	−0.017	0.520 (self)
Pythia $\times$ WikiText	10	0.623 (self)	0.557	0.582
Pythia $\times$ WikiText	20	0.051 (self)	0.196	0.222
Pythia $\times$ arxiv	10	0.352	0.751 (self)	1.144
Pythia $\times$ arxiv	20	0.230	0.378 (self)	0.177
Pythia $\times$ code	10	0.030	0.384	0.770 (self)
Pythia $\times$ code	20	−0.128	0.182	0.423 (self)
Gemma $\times$ WikiText	10	0.578 (self)	−0.282	−0.510
Gemma $\times$ WikiText	20	0.121 (self)	−0.797	−0.577
Gemma $\times$ arxiv	10	0.025	0.265 (self)	−0.024
Gemma $\times$ arxiv	20	−0.237	−0.015 (self)	−0.166
Gemma $\times$ code	10	−2.072	−2.241	0.347 (self)
Gemma $\times$ code	20	−1.759	−2.325	0.078 (self)

J. Mechanism, Faithfulness, and Agentic Readings

The main text frames the contribution as a measurement-validation result: the compositional utility gap of (Lee et al., 2025) has empirical content on architectural decompositions of trained transformers, in the specific sense that the metric generalises across data splits, varies across architectures, and distinguishes trained from parameter-scrambled controls. This appendix sketches the relationship between that result and four adjacent programs in philosophy of machine learning and mechanistic interpretability.

Reliability beyond accuracy. Grote et al. (2024) argue that an ML property claim earns epistemic standing relative to a null model: stability under intervention, generalisation across populations, robustness to confound, etc. Our 3×3 evidence grid instantiates that view at the metric-validation level rather than at the model-property level. The gap is a metric, not a model property; the validation question is whether the metric is responsive to the things it is supposed to be responsive to. The three axes (generalisation, architecture, training) are three reliability tests for the metric.

Causal abstraction. Geiger et al. (2025) ground mechanistic interpretability in a faithfulness condition: a high-level causal model is licensed when interventions on its variables predict counterfactuals at the low level. The compositional utility gap is not a faithfulness criterion; it is a partition-existence criterion. The two can in principle disagree: a partition can be log-pool-feasible without supporting interventionally valid high-level variables, and a partition can support interventionally valid variables without satisfying log-pool feasibility. Our validation shows the gap is non-trivial in its own right; whether it converges with faithfulness on the same partitions is the natural next study (Appendix I). Until that joint evaluation, the paper does not claim the gap is sufficient evidence for mechanistic explanation in Geiger’s sense; it claims only that the metric tracks something beyond architecture and initialisation.

Mechanism in philosophy of cognitive science. The mechanistic-explanation tradition of Craver (2007) and Bechtel (2008) characterises mechanistic explanation as the combination of *decomposition*, *composition*, *organisation*, and *activity*: identifying parts, showing how they combine, in what arrangement, and what each does. The log-pool feasibility criterion operationalises a necessary precondition for the decomposition step: when a candidate decomposition’s groups produce distributions that, log-pooled, do not lose information against the parent, the decomposition has at least the algebraic shape of a mechanistic part-attribution. Our contribution lies entirely in validating that precondition test: a positive gap, on its own, does not establish organisation or activity in Craver’s sense. We do not claim the network has mechanisms in the Craverian sense; we claim its log-pool decomposability is a non-trivial measurable property.

Mesa-optimisation and subagent discourse. The mesa-optimisation literature (Hubinger et al., 2019) reads a network’s parts as goal-directed inner computational agents. The compositional utility gap, as we use it, places a precondition on those readings rather than supporting or refuting them: an architectural subset that fails the gap test does not license further agentic claims, and an architectural subset that passes the test still requires substantial additional argument before agentic vocabulary becomes appropriate. The functionalist commitment we adopt is conservative (a part counts as a part only when it plays a functionally significant role in some account or discourse), and the gap, validated as a measurement tool, is one such account that the part-attribution claim must clear before agentic readings become available.

Three readings of what the metric tracks. Within the measurement-validation result, three interpretive readings remain compatible with the data and are not equivalent. First, the original framework’s *subagent* reading (Lee et al., 2025): feasible log-pool groups correspond to compositional latent agentic substructures, and the trained-vs-N1 contrast reflects learning of such substructures. Second, a more concrete *distributional* reading: the gap tracks a specific property of trained output distributions (e.g., redundancy, second-order coherence, or learned-margin geometry) that happens to be detectable via this particular log-pool functional but admits no agentic interpretation. Third, a more abstract *decomposition-coherence* reading: the metric identifies when architectural contributions can be aggregated into stable summaries without any commitment to whether those summaries refer to agents, computations, or representations at all. The empirical work in this paper does not adjudicate between these readings. It validates the metric as a measurement tool, which is the precondition for adjudication. The controls in Appendix I sketch how subsequent work could discriminate among them.